

C833

Chemistry with Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

An earth science teacher discussing limestone caves that took thousands of years to form asks what ion calcium (group 2) forms. Calcium is the fifth most common element in Earth's crust. Which ion is it?

- A. Ca^+
- B. Ca^{2+}
- C. Ca^{2-}
- D. Ca^{5+}

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A main-group element forms an ion with a 3– charge. In which group is it most likely found?

- A. Group 13
- B. Group 15
- C. Group 17
- D. Group 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A biology teacher, whose class is studying proteins in a unit that runs six weeks, asks which bond is most polar. The class keeps a model of a protein chain on the front table. Choose from C-H, N-H, O-H, and F-F.

- A. O-H
- B. N-H
- C. C-H
- D. F-F

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A science teacher writing up a fertilizer investigation for an eighth-grade unit needs the formula for calcium nitrate. Nitrate is NO_3^- . The investigation uses 12 bean plants. What is the formula?

- A. $\text{Ca}(\text{NO}_3)_2$
- B. Ca_2NO_3
- C. CaNO_3
- D. CaNO_{32}

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

5.

A physics teacher who also coaches the robotics team compares CO_2 and H_2O . Both molecules have polar bonds, and each is about 0.1 nm across. Which statement is correct?

- A. CO_2 is polar and H_2O is nonpolar.
- B. CO_2 is nonpolar because it is linear; H_2O is polar because it is bent.
- C. Neither is polar because they are the same size.
- D. Both are polar because both have polar bonds.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

6. [TRANSFER — reversed stem / different variable]

A molecule contains only polar bonds, yet the molecule as a whole is nonpolar. What must be true?

- A. Its shape is symmetric, so the bond dipoles cancel.
- B. Its bonds are actually nonpolar.
- C. It has lone pairs on the central atom.
- D. It must be ionic.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

7.

An earth science teacher notes that water (18 g/mol) boils at 100°C, while hydrogen sulfide (34 g/mol), found near hot springs that tourists visit, boils at about –60°C. Why does water boil so much higher?

- A. Molar mass always decides boiling point.
- B. Water's O-H bonds are stronger, so they take more heat to break.
- C. Water molecules form hydrogen bonds with each other.
- D. Water has the larger molar mass.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A teacher who writes practice sets in the evening asks: how many oxygen atoms are shown in 3Ca(OH)₂? Calcium hydroxide is used to treat acidic soil.

- A. 2
- B. 3
- C. 5
- D. 6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

How many formula units of Mg(OH)₂ contain a total of 10 hydrogen atoms?

- A. 10
- B. 2
- C. 5
- D. 20

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher on a field trip to a mountain camp at 3,000 m, where the group stayed four nights, heats water for cocoa and finds it boils at about 90°C. The pot is aluminum. Why?

- A. Higher air pressure on the mountain makes boiling easier.
- B. The aluminum pot conducts heat faster.
- C. The thermometer must be wrong, since water always boils at 100°C.
- D. Lower air pressure means the water's vapor pressure matches it at a lower temperature.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A pressure cooker cooks food faster than an open pot. What explains this?

- A.** The water boils away faster, concentrating heat.
- B.** Pressure itself cooks food, independent of temperature.
- C.** Higher pressure lowers the boiling point.
- D.** Higher pressure inside raises the boiling point, so the water gets hotter than 100°C.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

After a lab on crystal growing that 26 students enjoyed, a teacher has 500 mL of leftover copper(II) sulfate solution. The lab ran 15 minutes over. What should the teacher do with it?

- A.** Store it in an unlabeled jar for next year.
- B.** Pour it on the school garden, since copper is a plant nutrient.
- C.** Check the SDS and school policy, then place it in a labeled waste container for proper disposal.
- D.** Pour it down the sink with lots of water.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

An earth science teacher discussing limestone caves ~~that took thousands of years to form~~ asks what ion calcium (group 2) forms. ~~Calcium is the fifth most common element in Earth's crust.~~ Which ion is it?

Target word: ion charge **Decides:** Group 2 atoms lose two valence electrons.

A	Dropped element	Removes only one of the two valence electrons.
B	Correct	Losing two electrons gives a noble-gas configuration.
C	Inverted operation	Metals lose electrons; right size, wrong sign.
D	Camouflage	Uses the "fifth most common" detail, which does not decide the charge.

2. Answer B Concept 1 • transfer

A main-group element forms an ion with a 3[−] charge. In which group is it most likely found?

Target word: solve for group from charge **Decides:** Gaining 3 electrons reaches 8 from 5.

A	Inverted operation	Group 13 loses 3 electrons to form 3 ⁺ .
B	Correct	Five valence electrons + 3 = 8.
C	Confusion set	Halogens gain only one electron.
D	Wrong stem number	Reads the charge as the group number.

3. Answer A Concept 2

A biology teacher, whose class is studying proteins ~~in a unit that runs six weeks~~, asks which bond is most polar. ~~The class keeps a model of a protein chain on the front table.~~ Choose from C-H, N-H, O-H, and F-F.

Target word: most polar bond **Decides:** O-H has the largest electronegativity difference in the list.

A	Correct	Oxygen is much more electronegative than hydrogen (about 1.2 difference).
B	Adjacent category	Polar, but the N-H difference is smaller than O-H.
C	Inverted operation	The least polar bond in the list.
D	Confusion set	Fluorine is the most electronegative element, but two identical atoms share equally.

4. Answer A Concept 3

A science teacher writing up a fertilizer investigation ~~for an eighth-grade unit~~ needs the formula for calcium nitrate. Nitrate is NO₃[−]. ~~The investigation uses 12 bean plants.~~ What is the formula?

Target word: formula with a polyatomic ion **Decides:** Ca²⁺ needs two NO₃[−] units.

A	Correct	Two nitrate units balance +2; parentheses keep the ion intact.
B	Inverted operation	Puts the 2 on calcium instead of on nitrate.
C	Dropped element	One nitrate leaves a +1 charge.
D	Adjacent category	Without parentheses the subscript changes the nitrate ion itself.

5. Answer B Concept 4

A physics teacher ~~who also coaches the robotics team~~ compares CO₂ and H₂O. Both molecules have polar bonds, ~~and each is about 0.1 nm across~~. Which statement is correct?

Target word: which molecule is polar **Decides:** CO₂ is linear; H₂O is bent.

A	Inverted operation	Swaps the two.
B	Correct	Symmetry cancels CO ₂ 's dipoles; water's bent shape does not.
C	Camouflage	Size does not decide polarity.
D	Absolute language	Polar bonds can cancel in a symmetric shape.

6. Answer A Concept 4 • transfer

A molecule contains only polar bonds, yet the molecule as a whole is nonpolar. What must be true?

Target word: reverse from result to cause **Decides:** Only symmetry can cancel bond dipoles.

A	Correct	Examples: CO ₂ , CCl ₄ .
B	Inverted operation	Contradicts the stem.
C	Confusion set	Lone pairs usually break symmetry and make a molecule polar.
D	Adjacent category	Ionic compounds are not described as polar or nonpolar molecules.

7. Answer C Concept 5

An earth science teacher notes that water (18 g/mol) boils at 100°C, while hydrogen sulfide (34 g/mol), ~~found near hot springs that tourists visit~~, boils at about –60°C. Why does water boil so much higher?

Target word: explain the boiling points **Decides:** Water forms hydrogen bonds; H₂S does not.

A	Absolute language	Intermolecular force type can outweigh mass.
B	Adjacent category	Boiling separates molecules; it does not break bonds inside them.
C	Correct	H bonded to O allows strong hydrogen bonding.
D	Inverted operation	H ₂ S is heavier; mass alone predicts the opposite.

8. Answer D Concept 6

A teacher ~~who writes practice sets in the evening~~ asks: how many oxygen atoms are shown in 3Ca(OH)₂? ~~Calcium hydroxide is used to treat acidic soil.~~

Target word: count O atoms **Decides:** 3 × (1 × 2) = 6.

A	Dropped element	Ignores the coefficient 3.
B	Dropped element	Ignores the subscript 2 outside the parentheses.
C	Inverted operation	Adds the coefficient and subscript instead of multiplying.
D	Correct	Each formula unit has 2 O; three units have 6.

9. Answer C Concept 6 · transfer

How many formula units of $\text{Mg}(\text{OH})_2$ contain a total of 10 hydrogen atoms?

Target word: solve for the coefficient **Decides:** Each $\text{Mg}(\text{OH})_2$ has 2 H.

A	Wrong stem number	The atom count, not the unit count.
B	Confusion set	That is the H per unit, not the number of units.
C	Correct	$10 / 2 = 5$.
D	Inverted operation	Multiplies by 2 instead of dividing.

10. Answer D Concept 7

A teacher on a field trip to a mountain camp at 3,000 m, ~~where the group stayed four nights~~, heats water for cocoa and finds it boils at about 90°C . ~~The pot is aluminum~~. Why?

Target word: why the boiling point is lower **Decides:** Air pressure is lower at altitude.

A	Inverted operation	Pressure is lower, not higher, at altitude.
B	Camouflage	The pot changes how fast water heats, not the temperature at which it boils.
C	Absolute language	100°C holds only at standard sea-level pressure.
D	Correct	Boiling occurs when vapor pressure equals surrounding pressure.

11. Answer D Concept 7 · transfer

A pressure cooker cooks food faster than an open pot. What explains this?

Target word: reverse: higher pressure effect **Decides:** Higher pressure raises the boiling point.

A	Confusion set	The sealed cooker traps steam; the key is the higher temperature.
B	Adjacent category	Cooking speed comes from the higher temperature.
C	Inverted operation	Reverses the pressure effect.
D	Correct	Water must reach higher vapor pressure before it boils.

12. Answer C Concept 8

After a lab on crystal growing ~~that 26 students enjoyed~~, a teacher has 500 mL of leftover copper(II) sulfate solution. ~~The lab ran 15 minutes over~~. What should the teacher do with it?

Target word: disposal **Decides:** Copper compounds are hazardous waste; the SDS and school rules govern disposal.

A	Dropped element	Every container must be labeled.
B	Adjacent category	Trace-nutrient status does not make concentrated solutions safe to dump.
C	Correct	Heavy-metal solutions are collected, not poured out.
D	Confusion set	Dilution does not make copper waste safe for drains.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Valence electrons predict ion charge	Main-group atoms gain or lose electrons to reach a noble-gas count: groups 1, 2, 13 form +1, +2, +3; groups 15, 16, 17 form -3, -2, -1.	Giving an ion the charge equal to its number of valence electrons with the wrong sign.	A group number or valence-electron count and a question about the ion.
2	Electronegativity and bond polarity	The bigger the electronegativity difference between two atoms, the more polar the bond; identical atoms share equally.	Judging polarity by atom size or by how common the bond is.	A list of bonds and “most polar.”
3	Polyatomic ion formulas	Treat a polyatomic ion as a unit; use parentheses when more than one is needed, and never change the ion’s own subscripts.	Dropping parentheses or merging the ion’s subscript with the count.	Names ending in -ate or -ite, or ions like hydroxide and ammonium.
4	Molecular polarity depends on shape	A molecule with polar bonds is nonpolar if its shape is symmetric so the bond dipoles cancel.	Assuming polar bonds always make a polar molecule.	CO ₂ vs. H ₂ O, or “polar bonds but nonpolar molecule.”
5	Intermolecular forces and boiling point	Stronger attractions between molecules (hydrogen bonding > dipole-dipole > dispersion, for similar sizes) raise the boiling point.	Predicting boiling point from molar mass alone.	H bonded to N, O, or F compared with a heavier similar molecule.
6	Counting atoms in formulas and equations	Atoms of an element = coefficient × subscript (× the outside subscript when parentheses apply).	Ignoring the coefficient or the subscript outside parentheses.	A coefficient in front of a formula with parentheses.
7	Boiling point depends on pressure	A liquid boils when its vapor pressure equals the surrounding pressure, so lower air pressure means a lower boiling point.	Believing water always boils at 100°C.	Altitude, a vacuum chamber, or a pressure cooker.
8	Chemical labeling and waste disposal	Every container must be labeled; the Safety Data Sheet (SDS) and school rules decide storage and disposal, and hazardous waste goes into labeled waste containers, not the sink.	Pouring small amounts down the drain or using unlabeled chemicals.	Leftover solutions or an unlabeled bottle.

Course scope (public description): Chemistry with Lab for graduates provides already licensed teachers seeking an additional license or endorsement in middle grades science or secondary physics, biological science, or earth science with an introduction to the field of chemistry. Designed for those not majoring in chemistry education, this course highlights how the topics covered can be applied within various branches of science. This course provides students with opportunities to examine the electronic structure of atoms, study periodic trends, name chemical compounds, write chemical formulas, determine the structure of molecules, balance chemical reactions, and discover the changing states of matter. Laboratory experiences facilitate the study of matter and the application of laboratory safety and maintenance procedures. Concepts in Science for graduates is a prerequisite for this course.